



**Loyola Marymount
University**

Smart Car Supreme

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**Loyola Marymount University
Department of Electrical and Computer
Engineering
Senior Capstone 2025-2026**



Joshua Oliveira –
Computer Engineer
Autonomous Systems
Lead



Brayden Kimura –
Computer Engineer
Lane Detection Lead



Johnathan Yi –
Electrical Engineer
Hardware Design Lead



Amelia Wirick –
Electrical Engineer
PCB Design Lead

Background & Motivation

Autonomous Vehicles

- Increased reliance on autonomous control systems in industry
- Developing full-scale autonomous vehicles is expensive and complex
- This is fifth iteration of this project at LMU
- Aligns with growing demand for autonomous systems experience
- Reproducible platform for future LMU engineering students and research teams



Challenges

What issues have we experienced?

- Autonomous driving
 - Perception
 - Shadows
 - Sun Glare
 - Control
 - Latency
 - Actuation
 - Speed Control
- Power & Hardware
 - Deciphering the old wiring and previous CAN system (incorrect master node, etc)
 - How to power integrated system components (Jetson, Arduino, etc.)
 - PCB capabilities and limits

Project Planning and Design Choices



Smart Car Design

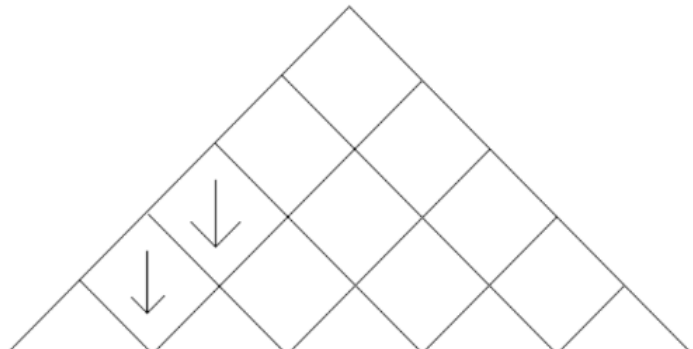
How Will it Work?

- Design a miniaturized autonomous RC car platform.
- Use PWM control for throttle and steering.
- Integrate camera and lane detection
- Employ custom PCBs and CAN bus communication for modularity.
- Develop scalable firmware for subsystem testing and future expansion.



Requirements

- Vehicle will stay in a designated lane within 6 inches
- Vehicle's maximum speed will be 3 feet per second
- Vehicle will have a minimum driving distance of 350 yards
- Vehicle will use multiple PCBs measuring at most 4 inches by 5 inches



		Max Speed	Stopping distance	Minimum Distance Covered	Use pcbs	Distance from lane
		+	+	+	+	+
Collision Avoidance	+	↓↓	↑↑			
Lane Keeping	+	↓				↑↑
Autonomously Drive Around Pereira Quad	+	↑	↑	↑↑		↑↑
safety	+	↓	↑↑			↑
Modular design	+				↑↑	
Cleaner wiring	+				↑↑	
Requirements	X	3 Ft/Sec	1 Ft	350Ft	Modular	6 In

Comparisons for Final Approach

Lane Detection

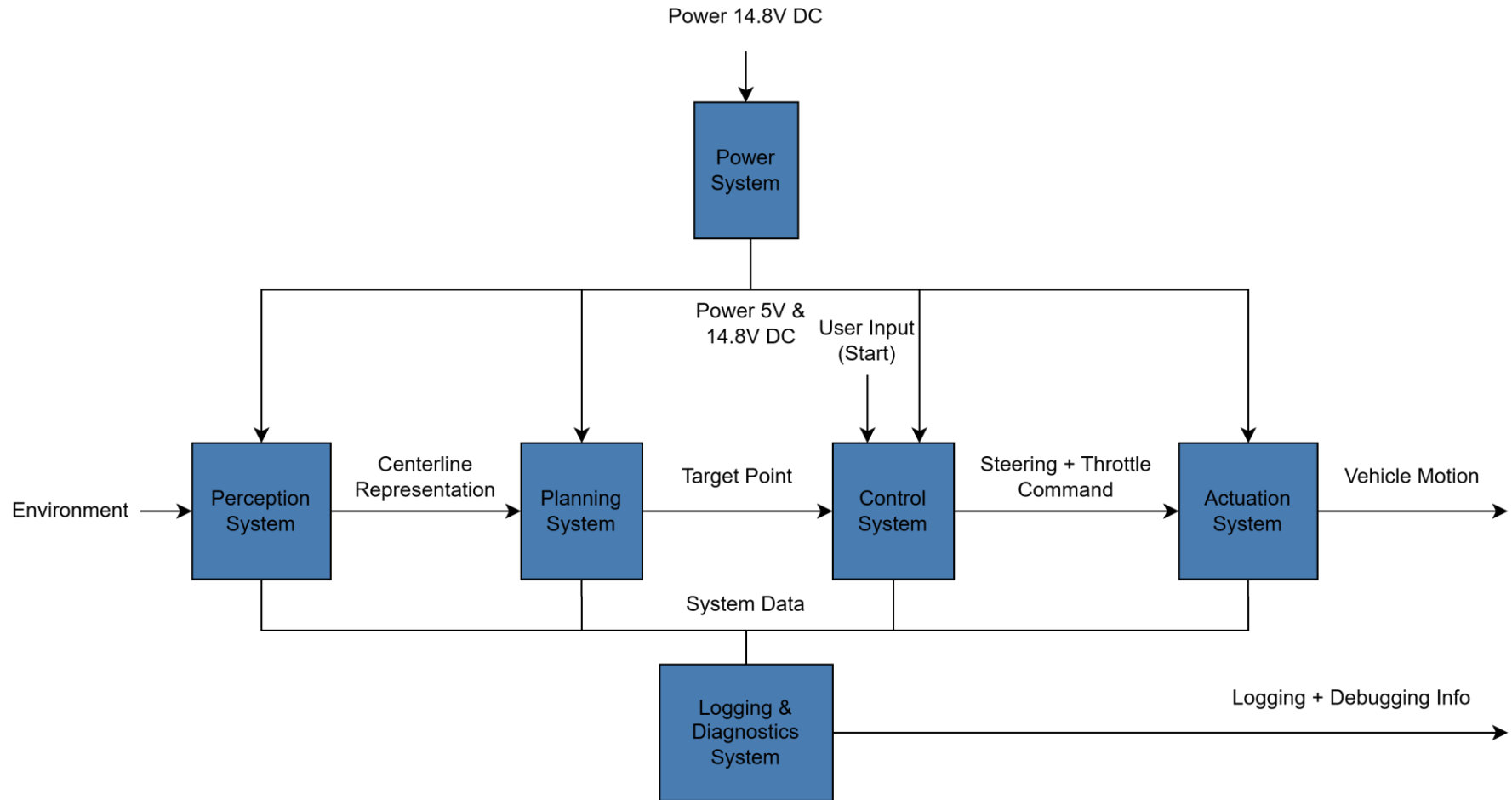
Technology	Advantages	Disadvantages
RGB camera	Low cost, ML-compatible, high resolution	Sensitive to glare and lighting
Depth camera	3D lane geometry	Expensive, less robust outdoors
LiDAR for lane detection	Excellent accuracy	Unnecessary for lane lines; high compute load
Hybrid LiDAR + camera	Strong perception	Too complex for RC-scale platform

Perception Platform

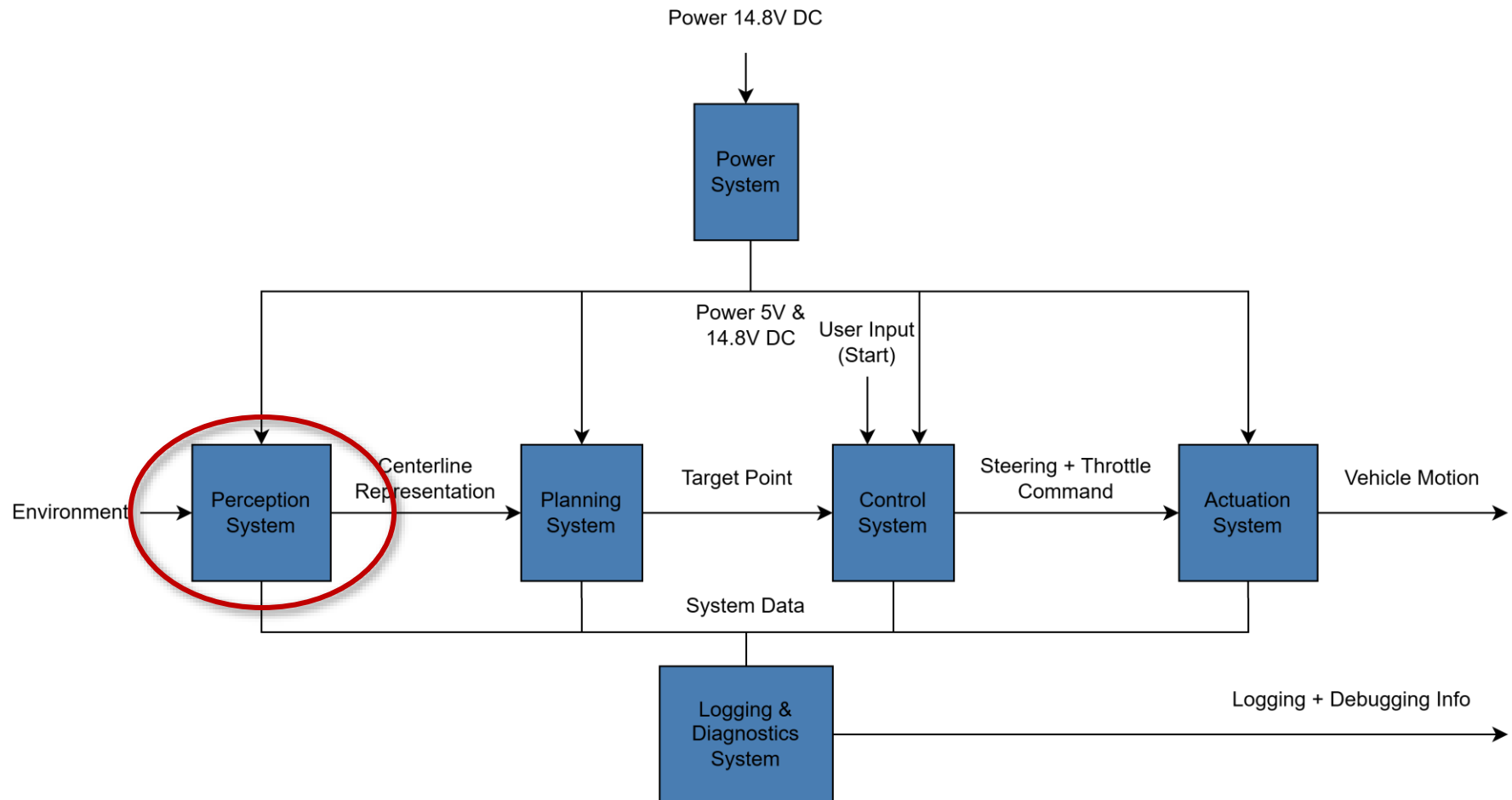
		Arduino Uno	Raspberry Pi 5	Jetson Nano	Jetson Orin
SDK/Driver Support	0.20				
I/O & Buses	0.15	0.15	0.2	0.3	0.35
Compute Power	0.25	0.42	0.18	0.21	0.19
Real-Time/Latency	0.11	0.14	0.28	0.28	0.3
Power & Thermals	0.09	0.28	0.28	0.28	0.16
Memory	0.08	0.25	0.25	0.25	0.25
Cost	0.04	0.08	0.18	0.44	0.3
Availability	0.04				
Form Factor	0.04	0.85	0.1	0.05	0
Score		0.164089	0.251758089	0.273079	0.311074



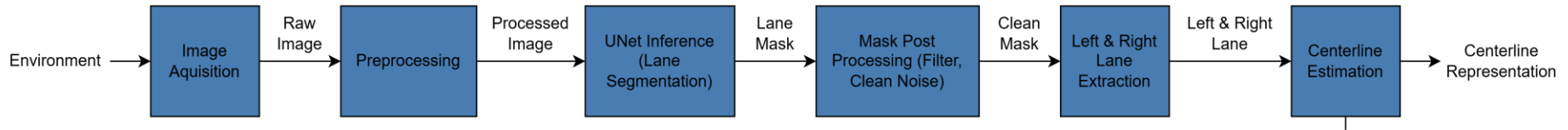
Functional Decomposition (Level 1)



Functional Decomposition (Level 1)



Functional Decomposition (Level 2)



Cost Table

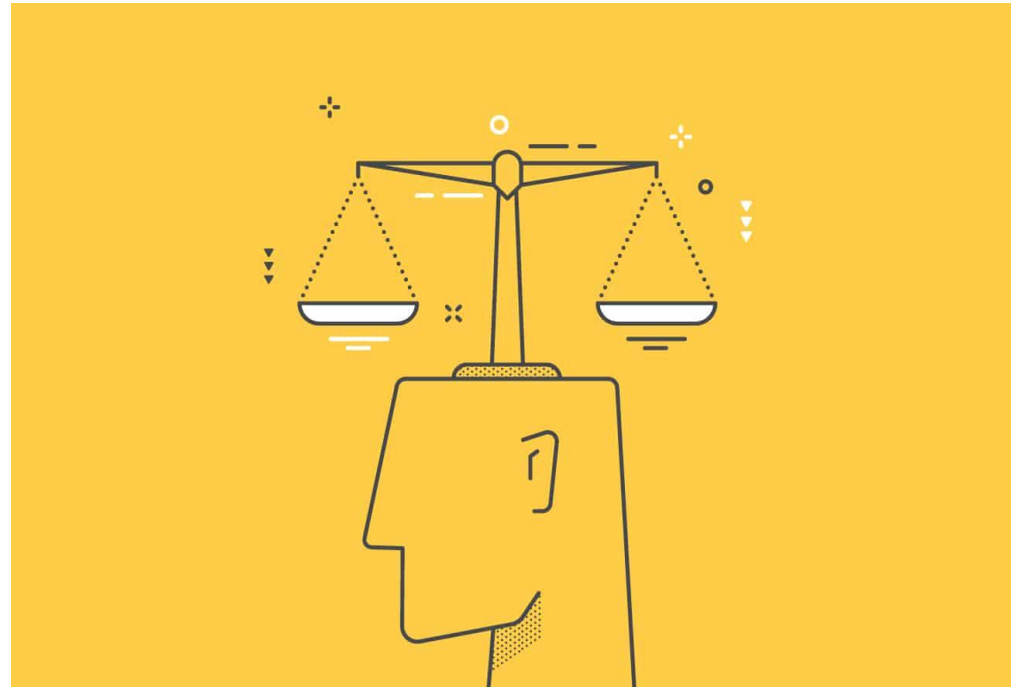
Item	Cost
RC Car	\$1200
Jetson Orin Nano Developer Kit	\$250
Arduino Uno	\$20
MCP2515	\$2
CAN Converter	\$30
Buck Converter	\$50
Camera	\$170
Ic3 to Ic5 Adaptor	\$6 x2
Ic3 Y-Splitter	\$15 x2
Total	\$212

Engineering Context and Constraints



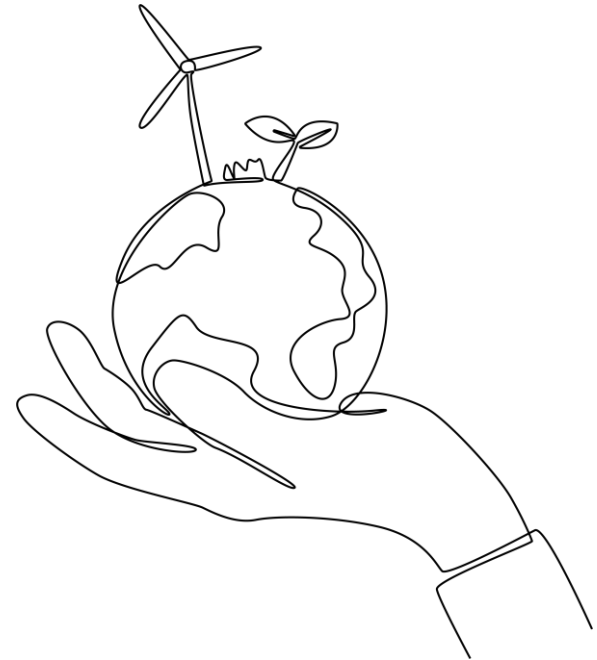
Ethics

- Safety features
 - Anti Collision
- Proper Documentation
- Potential negative impacts
 - Job Removal (taxi/uber)
 - Scale does not apply
- Potential benefits
 - Delivery usage
- Lack of human decision
 - How to handle accidents



Sustainable Development Goals

- 7: Affordable & Clean Energy
 - Electric Vehicle
- 8: Work & Economic Growth
 - Job Market Impact
 - Negative & positive
- 12: Responsible Consumption & Production
 - Modular PCB Design
 - Keeps design maintainable
 - Easy to revise



Standards



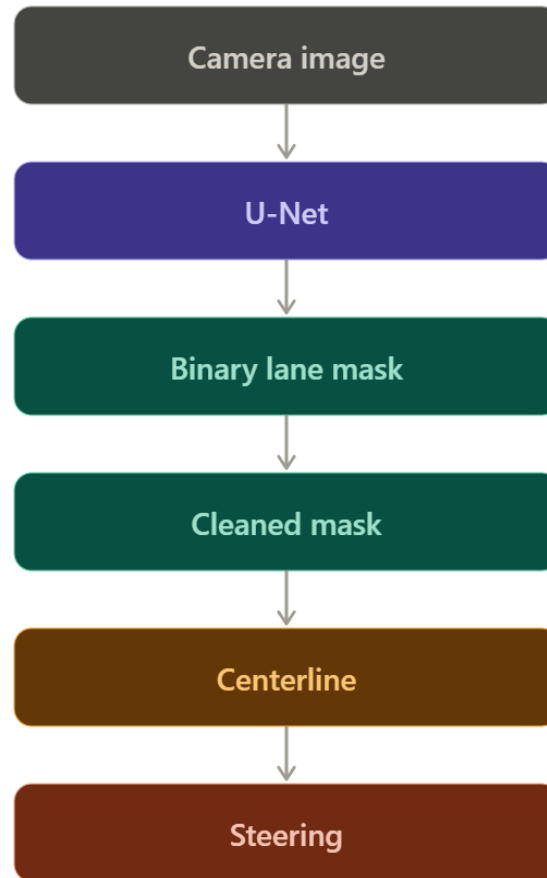
- IEEE 1722 — Time-sensitive transport for camera/sensor data
- IEEE 21451 — Smart sensor/transducer interface standard
- IEEE 29119 — Software testing & verification
- IEEE 1474 — Standards for intelligent transportation system control
- IEEE 1687 — PCB and embedded instrumentation access



Implementation and Results



Flowchart of SMART Car



Jetson

Brayden

- Brain of the car
- Runs U-Net lane detection model
- Finds centerline
- Sends steering and throttle commands via CAN



Arduino

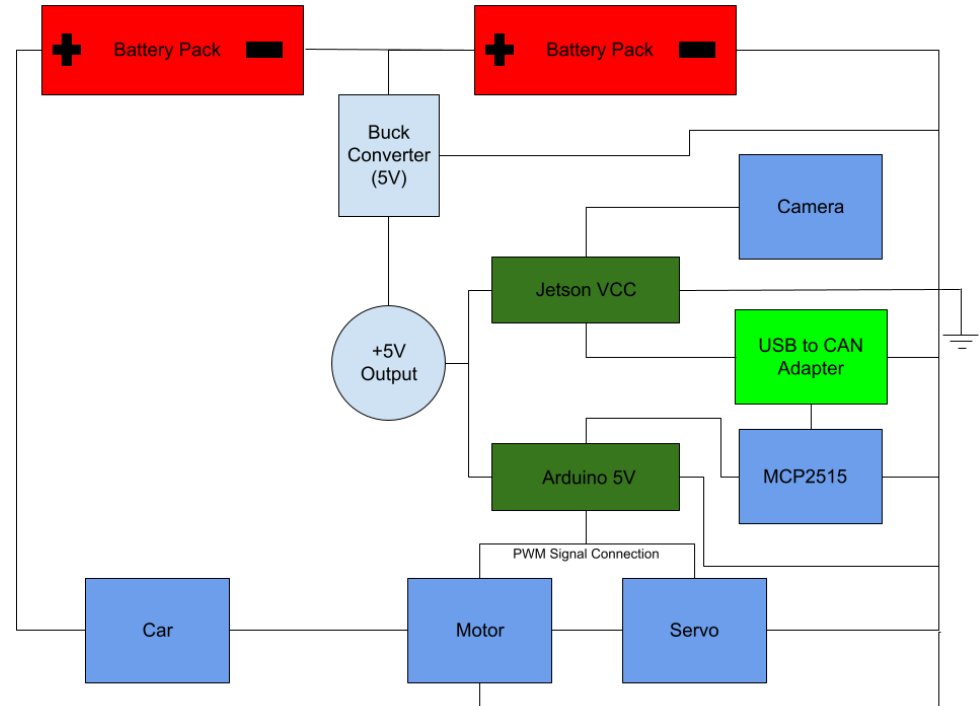
Brayden

- Arduino receives CAN data directly from the Jetson
- Ensures system is armed
- Outputs specific PWM signals to throttle and steering
- Safety feature sets car to neutral and disarms if it does not receive CAN messages from Jetson for longer than 250ms



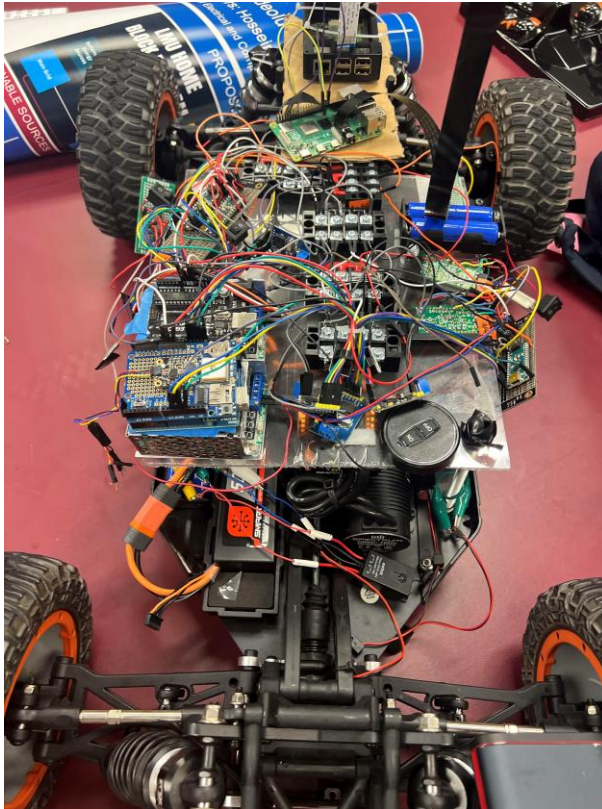
Hardware Integration

- Integrated extended hardware onto the platform powered by two LiPo batteries from car
 - First battery y-split to power embedded systems and car
 - Second battery directly goes to the car
 - Utilized IC3 to IC5 adaptors and IC3 Y-splitters
 - Jetson, Arduino, MCP2515, etc.

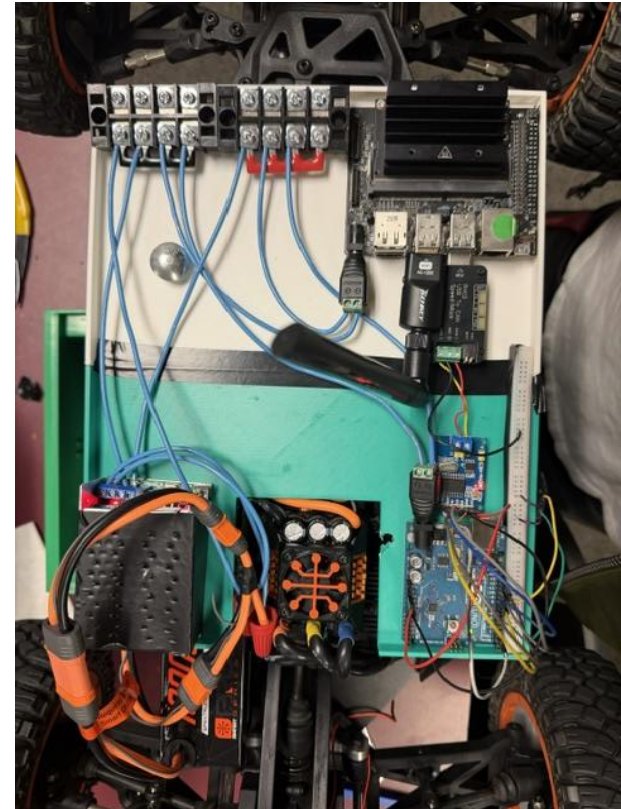


Hardware Integration

Joshua, Brayden, and Johnathan



Previous Version



Our Version

Self-Driving Program Pipeline

Joshua

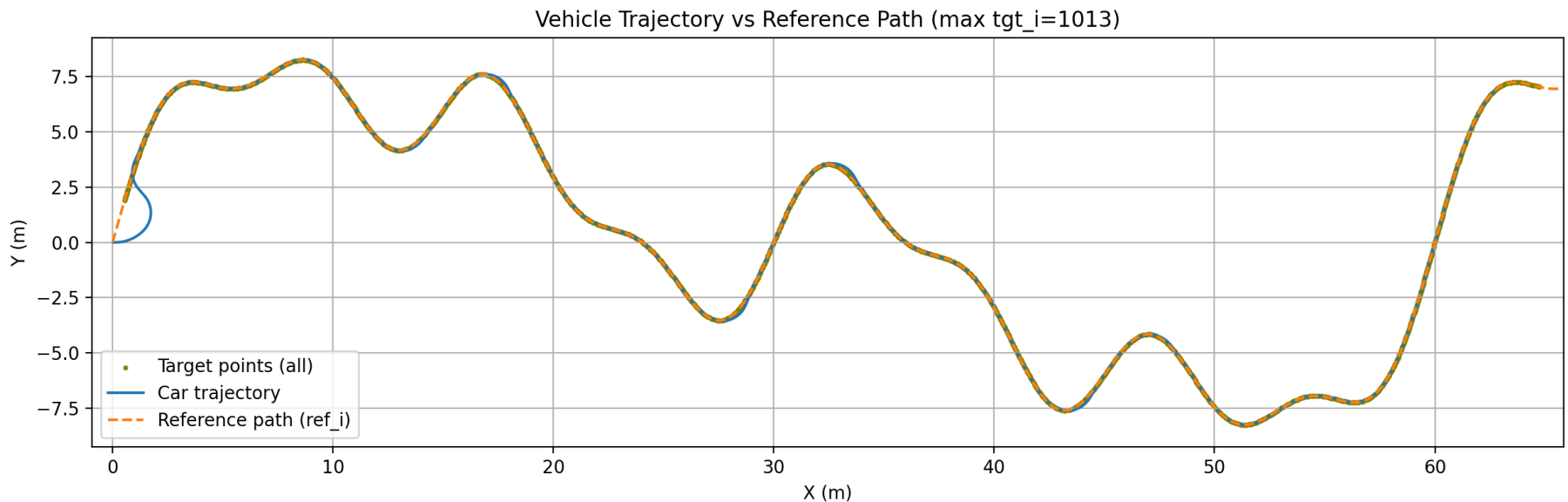
1. Lane Detection / Lane Input
 - Capture lane information from simulation or the forward-facing camera and produce centerline points representing the lane ahead.
2. Path Generation
 - Convert centerline points into a smooth, structured, ordered reference path that defines where the vehicle should drive.
3. Target Point Selection
 - Choose a lookahead point on the path relative to the car's current local frame.
4. Steering Control
 - Use Pure Pursuit to calculate the steering angle needed to drive toward the target point.
5. Safety & Limits
 - Apply steering limits, rate limits, and fallback logic when detection confidence is low or path data is unreliable.
6. Actuation
 - Convert steering and throttle commands into PWM values and send them to the vehicle through CAN.
7. Logging & Monitoring
 - Record lane data, target points, control commands, timing, and system status for debugging and performance analysis.



Self-Driving Program Simulated Test

Joshua

Graph showing simulated Self-Driving Program Stress Test



Lane Detection

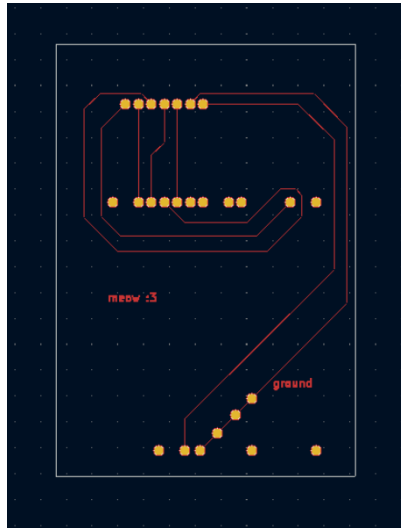
Joshua and Brayden



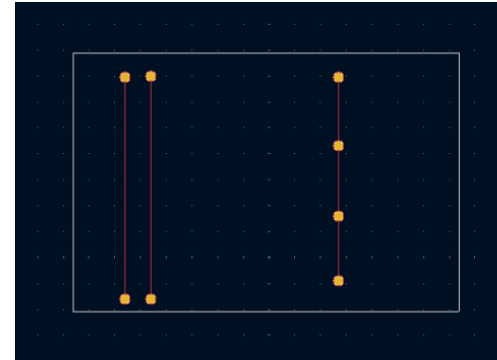
PCB Layouts

Amelia

- Designed two boards, one would connect Arduino and MCP2515 the other is for CAN Bus and ground



Arduino to MCP PCB Layout
Image Source: Senior Lab Laptop
Accessed: Apr 28, 2026

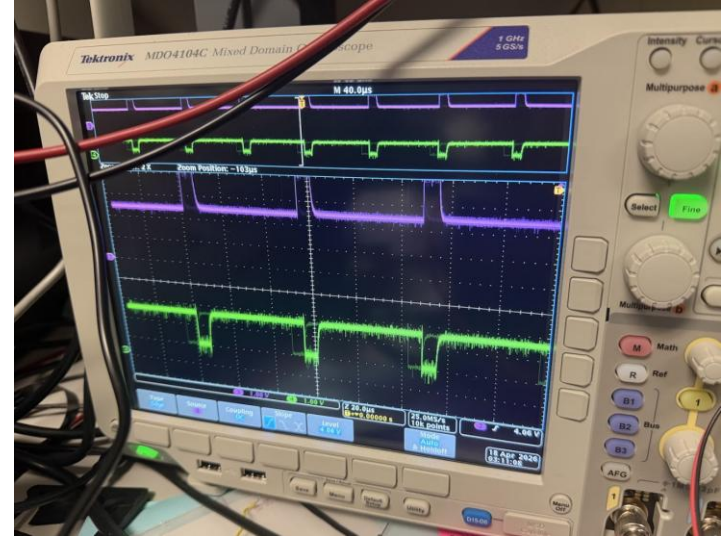


CAN Bus and ground PCB Layout
Image Source: Senior Lab Laptop
Accessed: Apr 28, 2026

PCB Off Car Testing

Amelia

- Tested the Arduino to MCP PCB not connected to the car initially
 - Used a program to send CAN signals between two Arduino using MCP2515s to convert to CAN
- CAN Bus portion of other PCB was also tested while not on the car using the oscilloscope



Oscilloscope

Image Source: Amelia Wirick's iPhone

Accessed: Apr 28, 2026

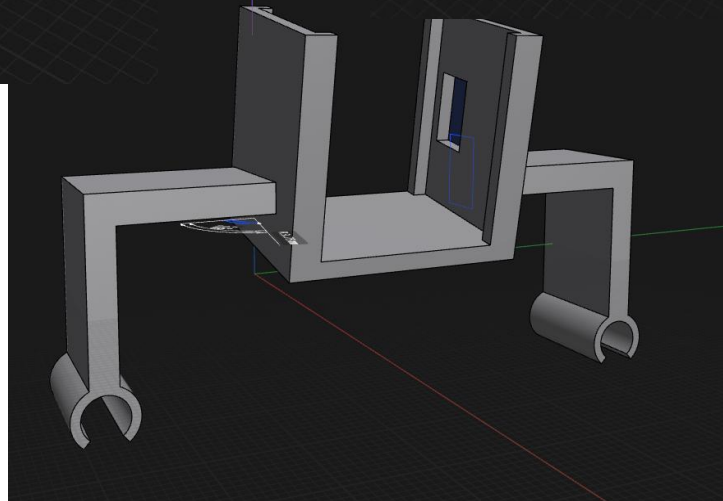
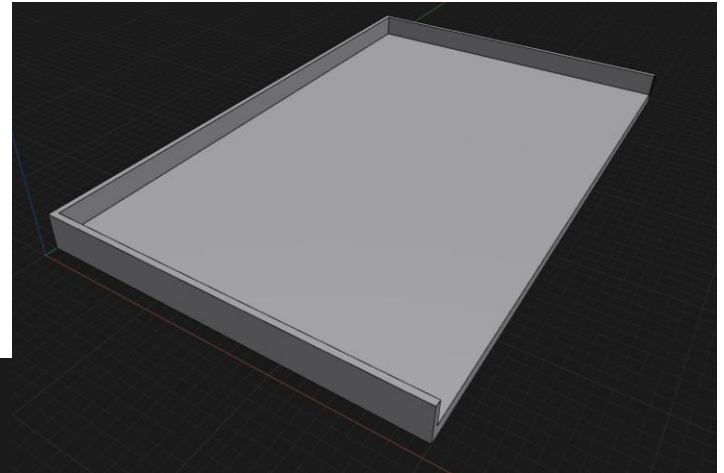
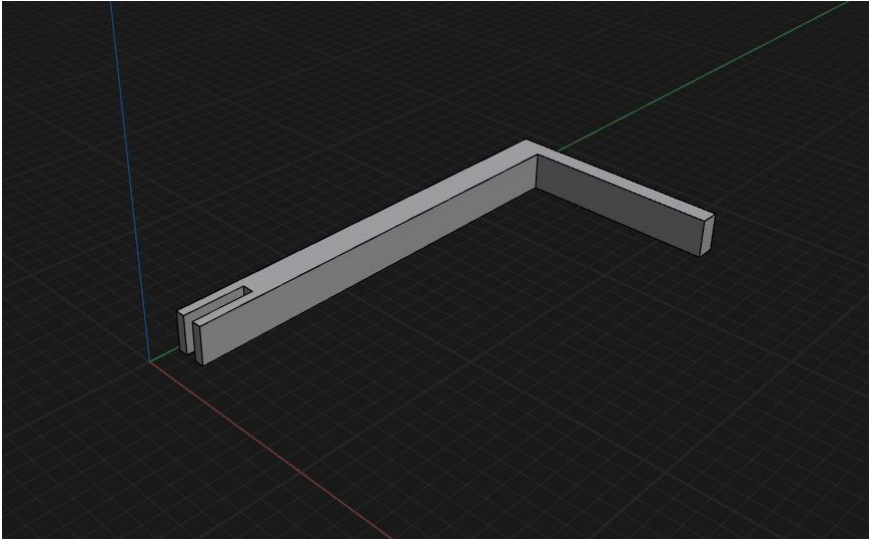
PCB On Car Testing

Amelia

- CAN Bus and ground PCB did not require much testing as the signals had been confirmed off car
 - Required soldering to connect
 - Originally tested CAN Bus and ground on separate boards
- Used a modular approach to test the Arduino to MCP PCB

CAD Screenshots

Johnathan



Conclusion



Final Benchmark

Amelia, Brayden, Johnathan, and Joshua

- Vehicle will not leave the designated lane within 6 inches
 - 4 inches
- Vehicle's maximum speed will be 3 feet per second
 - Slower than 3 feet per second
- Vehicle will be able to stop within a foot of any obstacle
 - LiDAR for obstacle avoidance was not integrated
- Vehicle will have a minimum driving distance of 350 yards
 - Vehicle has been tested around the full track
- The vehicle will use multiple PCBs measuring at most 4 inches by 5 inches
 - Vehicle will use one PCB



Future work

Amelia, Brayden, Johnathan, and Joshua

- Further PCB Integration
 - Connecting the Arduino to the MCP 2515
- Integrate LiDAR
 - Integrate obstacle avoidance
- Dynamic speed control



Acknowledgments

Amelia, Brayden, Johnathan, and Joshua

The SMART Car Supreme team would like to thank the following individuals for their assistance in the progress of this project:

- Dr. Hossein Asghari
- Leah Ghebrani
- LMU EECE Department



Questions?

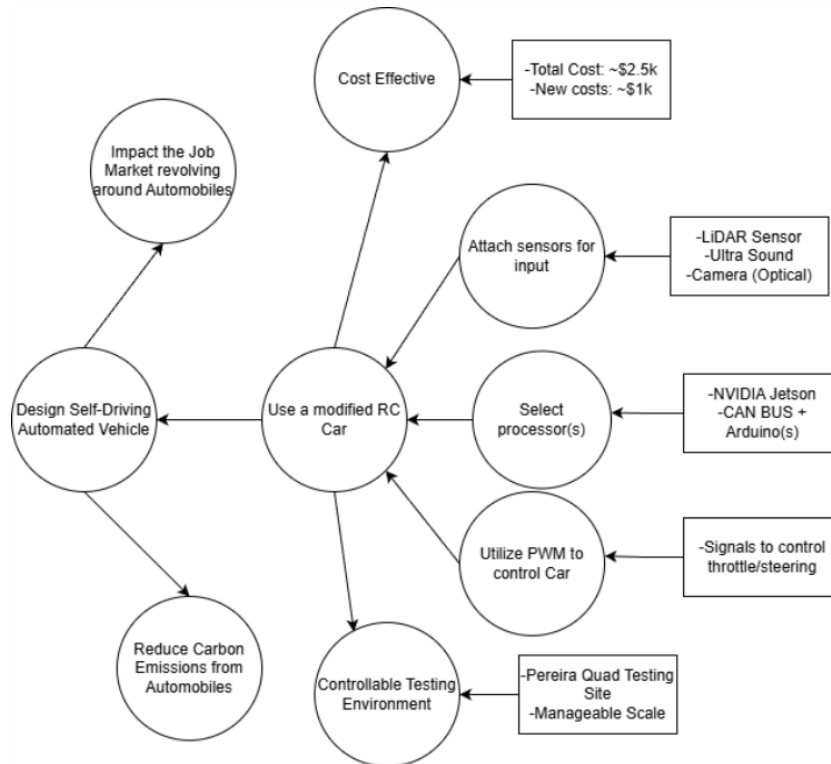


TEST VIDEO

Amelia, Brayden, Johnathan, and Joshua



Concept Generation



Considerations:

- Jetson vs Alternatives for compute platform
- Camera vs LiDAR for lane detection
- Breadboard vs PCB for electronics

Final selected architecture:

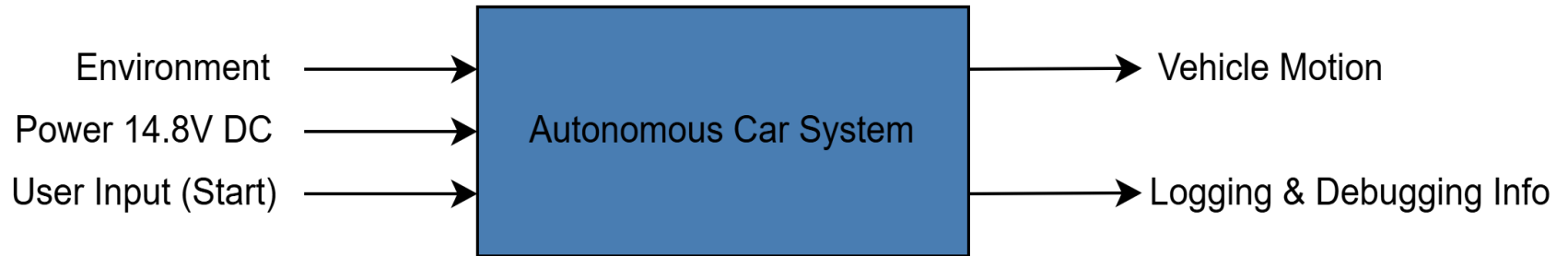
- Jetson Orin Nano for perception and decision-making
- Arduino CAN nodes for actuation
- High-dynamic-range GMSL2 camera for lane detection
- Custom PCBs for clean, modular, maintainable system wiring

Initial Test Plans

- Goal: Implement CAN Communication
- Steps:
 - Test if just 1s and 0s can be sent
 - Test that varied signals can be sent
 - Test PWM Signal output on predetermined controls
 - Test PWM Signal output with live controls
- Goal: Complete a lap around Pereira
- Steps:
 - Test the program on a straight segment
 - Test the program on a curved segment
 - Test the program on varied segments
 - Test the program on the full path
- Goal: Implement Lane Detection
- Steps:
 - Test Preprocessing
 - Test UNet Model
 - Test Postprocssing
 - Test Lane Extraction
 - Test Centerline Estimation
 - Test Full End-to-End
- Goal: Use PCBs to clean up wiring
- Steps:
 - Test off car
 - Component 1 to wire to board to wire to Component 2
 - Component 1 to board to wire to Component 2
 - Component 1 to board to Component 2
 - Test on car with same process



Functional Decomposition (Level 0)



Functional Decomposition (Level 0)

